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Incorporating strafing gain into redirected walking with pose score guidance☆

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ABSTRACT

Redirected walking (RDW) is a virtual reality locomotion technique that enables users to explore large virtual environments within a limited physical space. While state-of-the-art methods based on physical trajectory planning make effective use of physical space, some of them often compromise user comfort due to frequent directional reversals in curvature gain. To address this, this paper proposes a novel RDW method that integrates strafing gain with pose score guidance. Our approach discretizes the physical space into a series of standard poses, each with a long-term safety score, and redirects the user toward the optimal pose. The main contribution is a path generation algorithm that decomposes redirection into two sequential stages to ensure stable gains for each planned path: it first uses the curvature gain to steer the user along an arc for orientation alignment, and then inserts a straight path segment with constant strafing gain to achieve positional alignment with the target pose. Simulation experiments demonstrate a reduction in resets, while the user study shows lower Simulator Sickness Questionnaire scores compared to previous methods. Our work explores the potential of combining novel gains with state-of-the-art methods to create a more effective and comfortable RDW controller algorithm.

1. Introduction

Virtual Reality (VR) technology provides immersive and realistic content, owing to significant advancements in technologies such as three-dimensional content generation and wearable interactive devices. It is now widely integrated into a variety of domains, including education, healthcare, and entertainment. However, a major challenge for VR technology is overcoming the conflict between a vast virtual environment (VE) and the limited physical space (PE). To this end, a variety of virtual locomotion techniques have been proposed, such as teleportation [1], walking-in-place [2,3], omnidirectional treadmill [4], and Redirected Walking (RDW) [5,6]. Among these, RDW holds a unique advantage as it enables users to walk naturally, providing a higher sense of immersion and spatial presence.

Redirected Walking leverages the complexities of the human perceptual system to apply imperceptible manipulations to the virtual scene. This remapping of physical walking to virtual movement is known as redirection gains. To effectively implement these gains and maximize physical space utilization, researchers have developed various RDW controllers that calculate and apply gains according to the user's current state and environmental limitations, redirecting them away from physical boundaries. However, many widely used controllers, including Steer-to-Center (S2C) [7], Steer-to-Orbit (S2O) [7], and Artificial Potential Fields (APF) [8–10], derive their decisions exclusively from the user's present state, a limitation that often traps them in local optima and degrades their performance in complex physical environments.

In recent years, advanced physical trajectory planning methods have been introduced to address this limitation. Among this class of methods, optimal pose guided (OPG) [11] is a fundamental method and has outstanding performance in complex physical environments, as it simultaneously considers the user's current and future pose (defined by a position and an orientation). However, the transfer path generation in OPG primarily depends on iteratively inserting 'S'-turns to connect

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the current and target poses. These 'S'-turns require curvature gain to frequently switch from a maximum value in one direction to the maximum in the opposite, which can be more perceptible to users and potentially lead to discomfort such as dizziness.

Consequently, it is logical to explore whether the user's walking path can be adjusted without frequently steering their turn. A potential solution is strafing gain [12], a novel type of gain that applies a subtle lateral displacement as the user walks forward, causing them to move along a diagonal path while maintaining their intended orientation. However, effective integration of strafing gain into advanced RDW controllers remains an underexplored problem.

This paper proposes a novel RDW method that incorporates strafing gain to address a specific issue in existing trajectory planning methods, where curvature gain frequently switches between opposite directions, which can be more perceptible to users and potentially lead to discomfort. We adopt the method of optimal pose guidance, discretizing the physical space into a series of standard poses, each comprising a position and an orientation, and pre-calculating a score for each pose that represents its long-term safety. With the highest-scoring pose set as the target, we introduce an innovative path generation algorithm to redirect the user. This algorithm decomposes the complex task of pose alignment into two distinct stages: first, it utilizes the maximum curvature gain to guide the user along a single circular arc to achieve orientation alignment. Subsequently, at a suitable tangent point on the arc, it inserts a straight path applied constant strafing gain to align the user's position. This design ensures that the direction of applied gains remains stable within a single guidance maneuver, thereby effectively decreasing the number of resets while significantly enhancing user comfort.

The main contributions of the paper are as follows:

1. We propose a novel path generation algorithm that combines curvature and strafing gains to stably redirect the user to the target pose.
2. Based on this algorithm, we implement a novel RDW controller. To the best of our knowledge, this work is the first to integrate strafing gain into RDW controllers.
3. The results of simulation experiments demonstrate that our method reduces the number of resets, while the user study shows that it improves user comfort with lower SSQ scores.

2. Related work

2.1. Redirection gains

Redirected Walking [5] was first proposed by Razzaque et al. in 2001. It leverages the complexities of the human perceptual system to apply imperceptible manipulations to the virtual scene, thereby remapping the relationship between physical and virtual movements. This technique allows users to experience large-scale virtual locomotion within a limited physical space. The mapping between physical and virtual motion is defined by redirection gains, which were initially classified by Steinicke et al. [6] into three categories: translation gain scales the user's virtual movement, rotation gain alters the angle of the user rotations, and curvature gain bends straight paths into curves.

Subsequent research has defined and measured more novel types of gains. These include bending gain [13], a variant of curvature gain; deviation gain [14], which converts stationary rotations into translational shifts; bidirectional rotation difference (BiRD) gain [15], which uses the user back-and-forth head movements to deflect their path. Furthermore, research has extended into the vertical dimension, introducing slope gain [16,17], vertical gain [18], and jumping gain that manipulate jumping actions.

Early research generally avoided applying gains during non-forward movements like sideward walking. However, the work of Cho et al. [19] revealed that users are actually less sensitive to gains during non-forward motion, which results in wider perceptual thresholds. Building on this, Dong et al. [20] developed FREE-RDW, a method that integrates these thresholds into a multi-user collision avoidance strategy. Collectively, these studies explored the potential of lateral motion, establishing its viability for enhancing RDW without significantly increasing simulator sickness.

While the aforementioned studies apply gains to a user's lateral movements, the strafing gain proposed by You et al. [12] explores a different but related concept. It applies a lateral displacement to a forward-walking user, causing them to move along a diagonal path in the physical world while their virtual path remains straight (Fig. 1a). Unlike curvature gains that induce rotation, their study confirmed that users maintain their heading and respond to the lateral shift with a diagonal path, rather than a rotational adjustment. This characteristic gives strafing gain significant potential in various scenarios, such as subtly adjusting the user's physical position to avoid obstacles, combining with other gains for complex path planning, or assisting in aligning physical and virtual environments. However, although the perceptual thresholds for strafing gain have been measured, its effective integration into advanced RDW controllers remains a direction that requires further exploration.

2.2. RDW controller algorithms

Recent advances in RDW have pushed the field toward several challenging directions, aiming to overcome physical space limitations while providing more natural virtual experiences. As summarized in a recent comprehensive survey by Liu et al. [21], these directions include the development of more sophisticated controller algorithms, the creation of coordination strategies to support multi-user environments, the integration of multisensory cues, and the design of methods capable of adapting to dynamic physical spaces and obstacles, among others.

Among these, to effectively manage redirection gains and prevent users from colliding with physical boundaries, researchers have proposed a series of RDW algorithms. Razzaque proposed several algorithms in his pioneering work, including S2C, S2O, and Steer-to-Multiple-Targets (S2MT) [7], designed to guide users toward specific targets or regions within the physical space to minimize collisions. To handle more complex physical environments, the APF method is introduced to RDW, placing the user in a virtual force field of attractive and repulsive forces to guide them away from obstacles and toward open areas [8–10]. More recently, Chen et al. introduced the APF Steer-to-Target (APF-S2T) [22] to address the problem of traditional APF vectors in narrow corridors being directed toward opposing walls, instead guiding users toward more open spaces.

In recent years, Reinforcement Learning (RL) has provided a new direction for RDW algorithm design, differing from traditional rule-based methods. For instance, the Steer-to-Optimal-Target (S2OT) [23] method by Lee et al. uses RL to determine the optimal steering target; the Steering via Reinforcement Learning (SRL) [24] method by Strauss et al. utilizes RL to directly predict redirection gain values; and the Selective Redirection Controller (SRC) [25] by Lee et al. employs RL to dynamically switch between different controllers.

Concurrently, advanced methods based on physical trajectory planning have also been developed, which demonstrate superior performance in complex environments by evaluating the reachability of physical paths to optimize redirection decisions.

2.3. Physical trajectory planning for RDW

RDW algorithms based on physical trajectory planning address many constraints that previous methods could not handle by modeling and evaluating future physical paths. Xu et al. proposed optimal pose guided [11], which discretizes the physical space into a series of standard poses containing both position and orientation. It scores these

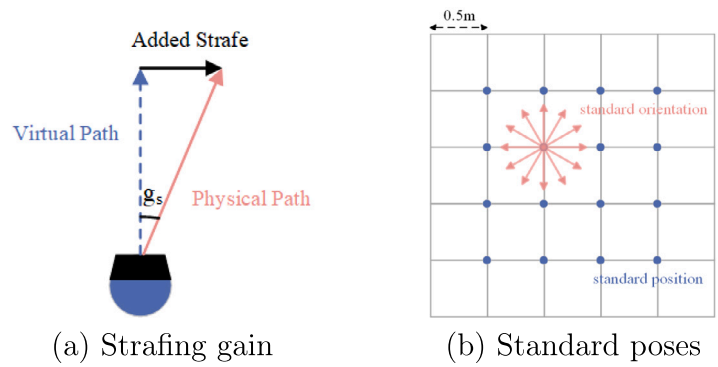


Fig. 1. Illustration of strafing gain and standard poses. (a) g_s refers to the angle to which a user is redirected. (b) The spacing between standard positions is set to 0.5 m and the number of standard orientations is set to 12.

poses based on the likelihood of their reachable paths colliding with obstacles and redirects the user to the optimal pose with the highest score among all reachable poses. However, its path generation algorithm, designed to connect poses, relies on iteratively inserting 'S'-turns, which may cause frequent changes in the direction of curvature gain and lead to user discomfort. POI-aware RDW [26] introduced a method that controls the location of resets to avoid them occurring near points of interest, which could disrupt the user immersive experience. It guides the user to any specified physical location within a reachable area by changing the steering direction only once per path. Fan et al. [27] proposed an RDW method based on user historical walking data to dynamically update the scores of different accessible poses in the physical space. SafeRDW [28] dynamically adjusts the physical path length to ensure that users are guided to a pre-defined safe area when jumping. Luo et al. [29] proposed an RDW method based on user intention prediction, which calculates a future virtual pose and aligns it with a physical pose to plan user trajectory.

We believe that there is still significant potential for innovation in redirection methods that use optimal pose guidance. Strafing gain could offer new possibilities for both algorithmic performance and user experience by subtly adjusting users physical position while maintaining their intended orientation. Therefore, this paper proposes a novel path generation algorithm that integrates strafing gain into physical trajectory planning, aiming to explore this potential to further enhance the performance and experience of RDW controllers.

3. Method

3.1. Pose discretization and scoring

Following the pose discretization and evaluation method of Xu et al. [11], we discretize the continuous physical space into a set of standard poses (Fig. 1b), each defined by a position and an orientation and denoted as $P = (x, y, \theta)$. Accessibility between any two standard poses is defined by the existence of a collision-free path that satisfies the gain thresholds, which is precomputed using a path generation algorithm detailed in Section 3.2. Based on the Bellman equation, the score $Q(P_i)$ for each standard pose P_i is calculated from the scores of all its accessible poses in the previous iteration, incorporating the distance to the target pose as a reward and resets as a penalty.

At runtime, the controller utilizes these precomputed scores to select a target pose with the highest score from all accessible poses according to the following formula:

$$P_{\text{target}} = \underset{P_i \in P_{\text{accessible}}}{\arg\max} \left(\text{dist}(P, P_i) + \gamma \cdot Q(P_i) - \eta \cdot reset\right) \tag{1}$$

Where $\text{dist}(P, P_i)$ represents the physical distance from the current pose to the accessible pose. $Q(P_i)$ is the precomputed score of the standard pose P_i, reflecting its long-term safety. γ is an attenuation coefficient. During the inference phase of the redirection controller, γ is set to 0.1 to avoid getting trapped in local optima, while during score precomputed phase it is set to 0.9 to enhance the impact of future reset penalties on the current decision. *reset* is a boolean variable indicating whether reaching P_i needs to be reset. If it does, a fixed penalty η is applied, with its value set to the area of the physical space.

3.2. Path generation algorithm

Given a starting pose and a target pose, our objective is to generate a smooth, collision-free path that satisfies the gain thresholds. To ensure the redirection remains imperceptible to users, our path generation algorithm applies only one of the two types of gain (curvature or strafing) one at a time.

As shown in Fig. 2, our approach first utilizes curvature gain to align the user orientation, followed by a path with only strafing gain, guiding users to the target pose without altering their intended orientation. Let the starting pose be P and the target pose be P^*. We denote the position of a pose as $\text{pos}(P)$ and its orientation as $\theta(P)$. A path is considered a set of continuous poses $\mathbb{C}$.

First, a circular arc path $\mathbb{C}_{\text{curv}}$ is generated from P, with its turning direction chosen to follow the shorter arc. To maximize the utilization of the physical space, we consistently apply the maximum gain that satisfies the curvature gain threshold. The path ends at an endpoint P' that satisfies the following:

$$\theta(P') = \theta(P^*) \tag{2}$$

The next step is to insert a strafing path $\mathbb{C}_{\text{straf}}$ into the arc path $\mathbb{C}_{\text{curv}}$ so that the final position P' matches that of the target P^*, while ensuring that the orientation of P' remains unchanged. Suppose the insertion point is P_i, at which the user's orientation is θ_i. Since the actual path with strafing gain is a straight line that deviates from the user's facing direction, it is necessary to modify the orientation of $\mathbb{C}_{\text{straf}}$ as follows:

$$\theta_{\text{straf}} = \theta_i + g_s \tag{3}$$

Where g_s is the value of strafing gain. To prevent abrupt changes caused by a mismatch between the two gain directions, we align its direction with that of the curvature gain. Similarly, to produce the maximum possible path offset, we prioritize using the maximum absolute value $|g_s|$ that satisfies the strafing gain threshold. After the insertion of $\mathbb{C}_{\text{straf}}$, the original arc path $\mathbb{C}_{\text{curv}}$ is split into two segments: an initial, unchanged segment and a second segment that is displaced by a vector shift. For the end P' of the path to coincide with the target P^*, the length and direction of $\mathbb{C}_{\text{straf}}$ must satisfy:

$$\begin{aligned} L_{\text{straf}} &= \|\boldsymbol{shift}\| = \text{dist}(\text{pos}(P'), \text{pos}(P^*)) \\ \theta_{\text{straf}} &= \arg(\boldsymbol{shift}) = \arg(\text{pos}(P^*) - \text{pos}(P')) \end{aligned} \tag{4}$$

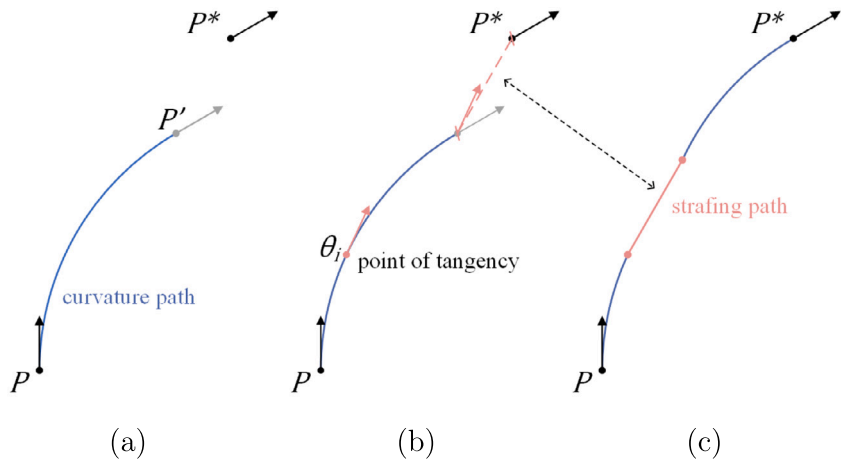


Fig. 2. Illustration of our path generation algorithm. (a) Applying curvature gain to steer the path until its final orientation at P' matches that of the target pose P^*. (b) The red dashed line represents the shift between P' and P^*, while the red solid arrow indicates the user actual direction θ_i. The insertion point is the point of tangency on the arc corresponding to θ_i. (c) Inserting the strafing path into the curvature path to make P' coincide with P^*.

Where $\boldsymbol{shift}$ represents the spatial offset required to get from P' to P*. Its magnitude determines the length of the strafing path, while its direction is that of the path itself. The required orientation θ_i is calculated using Eqs. (3) and (4). If this orientation falls within the angular range of the arc, $\theta_i \in [\theta(P), \theta(P')]$, then the continuity of $\mathbb{C}_{\text{curv}}$ guarantees that a corresponding insertion point P_i exists, and its position and orientation are calculated as follows:

$$(x_i, y_i) = \text{pos}(P) \pm R \begin{pmatrix} \sin(\theta_i) - \sin(\theta(P)) \\ \cos(\theta(P)) - \cos(\theta_i) \end{pmatrix} \tag{5}$$

$$P_i = (x_i, y_i, \theta_i) \tag{6}$$

Where R is the minimum arc radius that satisfies the curvature gain threshold, and the $\pm$ sign is determined by the turning direction of $\mathbb{C}_{\text{curv}}$. Although Eqs. (5) and (6) provide the theoretical coordinates of P_i, in our discretized implementation (as shown in Algorithm 1), we determine the insertion point P_i by iterating through all poses in $\mathbb{C}_{\text{curv}}$ to find the one whose orientation is closest to the target orientation θ_i.

If the condition for $\theta_i \in [\theta(P), \theta(P')]$ is not met, an attempt can be made to set $\theta_i = \theta(P)$ and solve for smaller absolute gain value $|g_s|$. If a valid value for g_s can be found, then the initial pose P is designated as the insertion point. If no path exists, or if the finally generated path $\mathbb{C}$ collides with boundaries or obstacles, the path from pose P to P^* is considered non-existent. When a path exists, it is denoted as $\mathbb{C}(P, P^*)$. The complete implementation process of the path generation described above is detailed in Algorithm 1.

3.3. RDW controller strategy

At runtime, our primary goal is to guide the user toward a standard pose with the highest long-term safety score. When the user walks forward, the controller first selects an accessible standard pose with the highest score as the target and generates a guidance path based on the algorithm in Section 3.2. For implementation purposes, this path is discretized into a set of path points at 0.1-meter intervals, where each point contains information on position, orientation, and the corresponding gain (i.e., whether to apply curvature or strafing gain, and its direction and value). During locomotion, in each frame, the controller matches the user current state to the nearest point on the path and applies the gain associated with that point, thereby ensuring the user follows the planned path under the gain constraints. To timely adjust the guidance path, we set thresholds. When the user's distance or orientation relative to the nearest point exceeds the thresholds, we consider the user rotates or deviates from the path, and the controller re-selects a target pose and generates a new path.

Algorithm 1: Path Generation Algorithm

Data: P, P^*
Result: path $\mathbb{C}(P_0, P_1, \ldots, P_n)$ from P to P^* which satisfies gains threshold

/* step 1: generate curvature path */
if $\theta(P) \neq \theta(P^*)$ **then**
  $\mathbb{C}_{curv} \leftarrow \langle P_0, P_1, \ldots, P_n \rangle$;
  /* with a Δs step size, sample a curvature path that starts at P and ends once its orientation matches P^*. */
  $P_0 = P, \theta(P_n) = \theta(P^*)$;
else
  $\mathbb{C}_{curv} \leftarrow \langle P \rangle$;
  /* curvature path not required. */
end
$\vec{shift} \leftarrow \vec{P_n P^*}$;
if $\|\vec{shift}\| < \epsilon$ **then**
  return $\mathbb{C}_{curv}$;
end
/* step 2: insert strafing path */
$\theta_{straf} = \theta(\vec{shift}), L_{straf} = \|\vec{shift}\|$;
if θ_{straf} *is in* $[\theta(P_0), \theta(P_0) + STRAF_MAX]$ **then**
  $i \leftarrow 0$;
  $g_s \leftarrow \theta_{straf} - \theta(P_0)$;
else if θ_{straf} *is in* $[\theta(P_0) + STRAF_MAX, \theta(P_n) + STRAF_MAX]$ **then**
  $g_s \leftarrow STRAF_MAX$;
  $i \leftarrow index(P_i)$;
  /* find pose in C_{curv} where $\theta(P_i)$ is closest to $\theta_{straf} - g_s$. */
else
  return no path found;
end
$\mathbb{C}_{straf} \leftarrow \langle P_{i+1}, P_{i+2}, \ldots, P_{i+k} \rangle$;
/* generate the straight-line path of length L_{straf} with strafing gain g_s sampled at a step size of Δs. */
$\mathbb{C} \leftarrow \langle P_0, P_1, \ldots, P_i \rangle \oplus \mathbb{C}_{straf} \oplus \langle P'_{i+k+1}, P'_{i+k+2}, \ldots, P'_{n+k} \rangle$;
/* P' denotes the post-path poses, translated to follow $\mathbb{C}_{straf}$. */

Rotation gain is applied when the user rotates. The controller selects an accessible standard pose with the highest score from the current position and sets the initial orientation of its guidance path as the target orientation. If the user turns toward the target orientation, maximum rotation gain is applied to encourage the turn; conversely, minimum

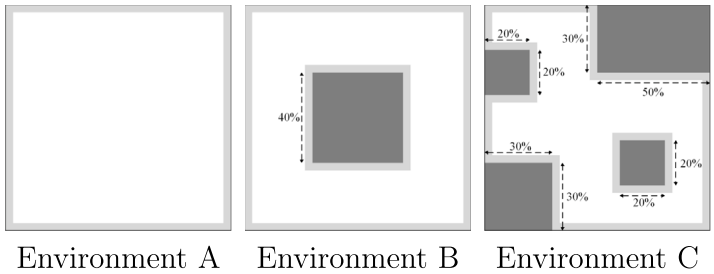


Fig. 3. Environment layouts used for simulation. Environment A is a control environment with no physical obstacles. Environment B is a moderately complex environment with a single square obstacle in the center. Environment C is a highly complex environment that simulates a real indoor room layout. We set a 0.5 meter buffer zone for obstacles and boundaries.

rotation gain is applied to discourage turning away. We consider standard poses that can only be reached via a reset as unreachable. When the controller cannot generate a path that requires no resets under the gain constraints, the system will only apply translation gain to help the user exit the current state quickly, after which regular path guidance is resumed.

The reset strategy adopts a traditional method similar to reset to gradient (R2G) [10]. When the user enters the buffer zone of a physical boundary, the controller selects a standard pose with the highest score that is reachable after a reset from the user's current position. It then generates a path and resets the user to the initial orientation of this new path.

4. Simulation experiment

4.1. Simulation design

Although research [30] indicates that generated paths may not accurately reflect the natural walking patterns of real users, simulation is still considered an effective way to evaluate RDW methods [31], as it allows a large number of tests to be conducted in various environments. As a preliminary evaluation of our proposed algorithm, we first conducted a series of simulation experiments with the OpenRDW platform [32]. We did not make any modifications to the source code of the existing baseline methods provided within the platform and implemented our algorithm as a new redirector by extending the base Redirector class. All experiments used the default gain parameters.

In the simulation experiments, a simulated user walks along predefined virtual paths. Upon reaching a waypoint, the user turns to face the next waypoint and then walks straight toward it. We generate 100 independent random virtual paths, each with a total length of 400 m. Waypoints are randomly generated within a range of 5 to 15 m from the previous one. To ensure fairness, all algorithms are tested on the same set of virtual paths.

We evaluated the RDW algorithms in three physical environments with different obstacle layouts, as shown in Fig. 3, representing a range of application scenarios from simple to complex. The obstacle-free Environment A represents the most common and ideal layout, applicable to scenarios where users have a clear, open tracking space. Environments B and C simulate some indoor physical settings, such as a furnished room, in order to test an algorithm's performance in reducing the spatial restrictions imposed on the user. The motivation for including these layouts is to explore the potential of RDW when a large, clear area is not available. Furthermore, this set of physical layouts is similar to prior work, allowing for a fair and direct comparison. All three layouts are tested in two sizes of square physical environments: 10 m × 10 m and 15 m × 15 m, resulting in a total of six simulated PEs. We set the virtual space as an obstacle-free open area, allowing the simulated user to walk freely within it.

Table 1
Post hoc pairwise comparisons of the number of resets between our method and other methods, showing adjusted p-values.

		S2C	S2O	TAPF	SRL	OPG
Env A	10 m	<.001***	<.001***	0.081	<.001***	0.005**
	15 m	<.001***	<.001***	0.036*	0.009**	0.405
Env B	10 m	<.001***	<.001***	<.001***	<.001***	0.635
	15 m	<.001***	1.000	<.001***	<.001***	0.004**
Env C	10 m	<.001***	<.001***	<.001***	<.001***	<.001***
	15 m	<.001***	<.001***	<.001***	<.001***	0.042*

Note: Significance levels are denoted by asterisks: * $p \leq .05$, ** $p \leq .01$, *** $p \leq .001$.

Table 2
Post hoc pairwise comparisons of average virtual distance between resets between our method and other methods, showing adjusted p-values.

		S2C	S2O	TAPF	SRL	OPG
Env A	10 m	<.001***	<.001***	0.013*	<.001***	<.001***
	15 m	0.089	<.001***	0.140	<.001***	0.590
Env B	10 m	<.001***	<.001***	<.001***	<.001***	1.000
	15 m	<.001***	<.001***	<.001***	<.001***	<.001***
Env C	15 m	<.001***	<.001***	<.001***	<.001***	0.780
	15 m	<.001***	<.001***	<.001***	<.001***	0.008**

4.2. Results and discussion

Fig. 4 presents a performance comparison of our algorithm (Ours) with five state-of-the-art RDW methods including S2C [7], S2O [7], TAPF [10], SRL [24], and OPG [11], across three physical environments (Environments A, B, and C) and two sizes (10 m × 10 m and 15 m × 15 m). For methods lacking a reset strategy, such as S2C and S2O, we employ two-one-turn as the default reset strategy. We analyzed the number of resets and average virtual distance between resets in all six environments.

As Shapiro–Wilk tests showed that not all results followed normal distribution, we employed the non-parametric Kruskal–Wallis H test to analyze the overall differences among the algorithm groups, followed by Dunn's post hoc test with Bonferroni correction to specifically compare the performance between our method and the others. Statistical analysis revealed statistically significant differences in the number of resets and average virtual distance between resets among the algorithms in all six environments ($p < .001$). Post hoc pairwise comparisons between our method and other methods are shown in Tables 1 and 2. Statistically, our algorithm demonstrated a significant improvement over all other algorithms except OPG in most tested environments. The only non-significant comparisons are with TAPF in Environment A at both 10 m × 10 m, and with S2O in Environment B at 15 m × 15 m. Compared to OPG, our algorithm was better at reducing resets in four environments except for the 10 m × 10 m Environment A and 15 m × 15 m Environment B.

Analysis of the experimental results demonstrates certain advantages of our method and the specific applicable scenarios for different

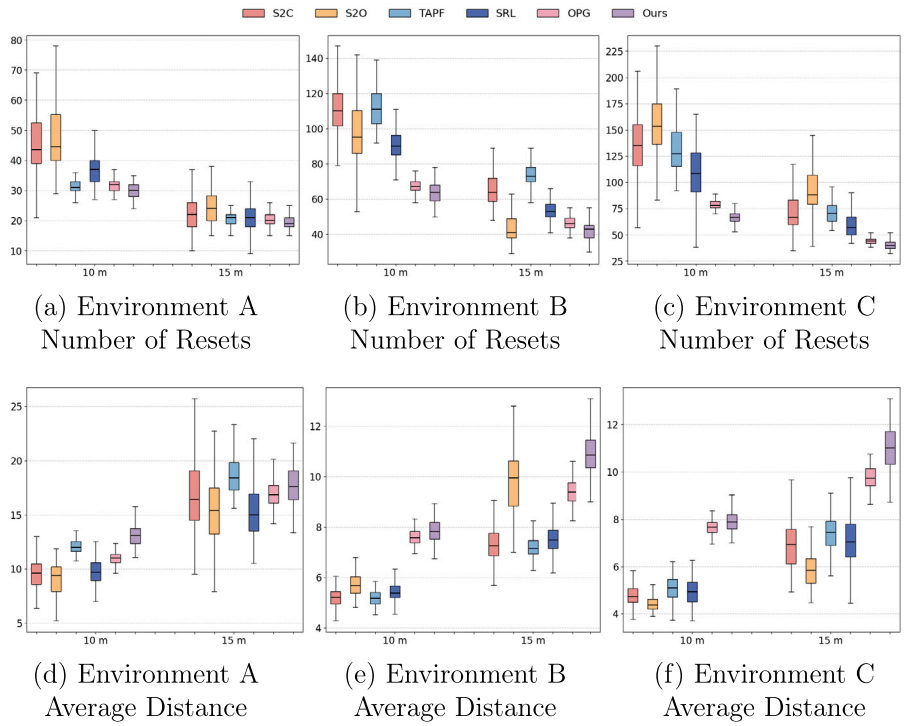


Fig. 4. The box plots of the results of experiments in the six environment-size combinations.

algorithms. For obstacle-free space like Environment A, our method's performance is comparable to several other algorithms and particularly shows no statistically advantage over TAPF, as APF-based methods can effectively utilize various gains based on the user's current position. For the more complex Environments B and C, our method and OPG hold a advantage. This is because these two methods guide users to the safest accessible future pose, helping them to effectively avoid obstacles and better utilize the physical space. The one exception is S2O in the 15 m × 15 m Environment B, as its steering orbit is designed to specifically bypass the central obstacle. Compared to OPG, the integration of strafing gain in our method enhances path planning flexibility while reducing computational complexity. These, combined with the modified rotation gain and reset strategy, allow our method to outperform OPG in most environments. It should be noted that while these differences are statistically significant, their practical impact may be more modest.

Table 3
Post hoc pairwise comparisons of the number of resets, average distance between resets, and SSQ scores between our method and other two methods, showing adjusted p-values.

	TAPF	OPG
Number of Resets	0.626	0.002**
Avg Distance	0.069	<.001***
SSQ Score	0.023*	0.019*

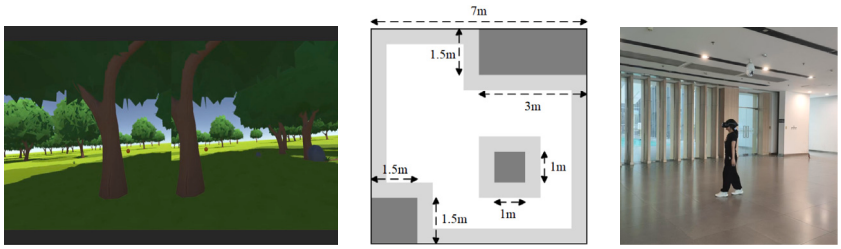
5. User study

5.1. User study design

To further evaluate the effectiveness and subjective comfort of our method, we conducted a user study following the simulation experiments. We recruited 23 participants (13 male, 10 female, aged 22–26), all of whom were right-handed, with 17 stating they had VR experiences. The experiment used an HTC VIVE Focus Vision head-mounted display (HMD), connected via wireless streaming to a laptop equipped with an AMD Ryzen 7 5800H processor, 16 GB of RAM, and an NVIDIA GeForce RTX 3060 GPU. Similar to the simulation, we conducted user study experiments using OpenRDW.

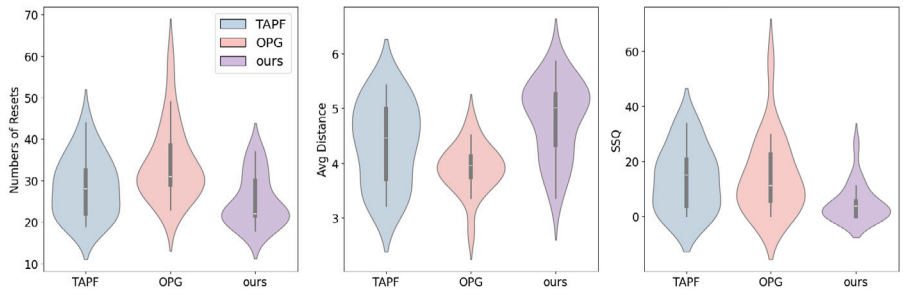
To ensure a stable and low-latency wireless stream between the HMD and the host computer, the physical tracking space is a 7 m × 7 m area. While the layout is a simplified version of Environment C with a 0.5 m buffer zone as shown in Fig. 5b, no actual obstacles were placed in the area to ensure the safety of participants. The virtual scene is an open forest environment (Fig. 5a), following the experimental setup of previous work [29]. The participants task is to collect orange spheres, which served as waypoints, in the virtual scene. At the start of each trial, the initial waypoint is created at a distance ranging from 2 to 8 m from the participant's current location, with a random angle between $-\pi$ and π radians. Upon reaching the waypoint, it disappears, and a new one is immediately generated based on the participant's current position. This process continues until a cumulative virtual distance of 80 m is reached for each trial.

We selected three RDW controller algorithms for the user study: TAPF, OPG, and our proposed method, as they demonstrated the best performance in the preceding simulation experiments. To ensure fairness, the same sequence of random waypoints was reused for each participant across the three conditions, while different participants were assigned different random seeds. Before the experiment began, participants filled out a pre-exposure Simulator Sickness Questionnaire (SSQ) [33]. A short training session with no redirection gains was then conducted to demonstrate task rules. Each participant completed three experimental sessions in an order counterbalanced by a Latin Square design [34]. Each session lasted approximately 7–8 min and was followed by a 5-minute break, during which participants filled out the SSQ again to assess their level of simulator sickness. The system automatically recorded the number of resets and the average virtual distance between resets. After all three sessions were completed, a brief interview was conducted to collect subjective feedback on walking naturalness and comfort. The entire experiment lasted approximately 40 min per participant.



(a)Vision field of both eyes (b)Physical layout (c)Test site photo

Fig. 5. The user study.



(a) Number of Resets (b) Average Distance (c) SSQ Scores

Fig. 6. Violin plot of the number of resets, average distance between resets, and SSQ scores in the user study.

5.2. Results

Fig. 6 shows a comparison of our method with TAPF and OPG on the number of resets, the average distance between resets, and SSQ scores. As some of the data failed the Shapiro–Wilk normality test, we used the non-parametric Friedman test to compare the three methods, showing that there are statistically significant differences in the number of resets (p = .001), the average distance between resets (p = .001), and SSQ scores (p = .021). The results of the post hoc analysis using the Wilcoxon signed-rank test with Bonferroni correction are detailed in Table 3.

As the violin plot and Table 3 show, the difference in the number of resets between our method and TAPF is not statistically significant. However, while our method performed slightly better than OPG in simulation environments, it showed a clear improvement in the user study. A possible reason is that real users do not move with perfectly smooth translation as in simulations. Their natural stride length may be too long to follow the short 'S'-turns in the path planned by OPG, which results in difficulty in replicating the simulated performance. In contrast, our method provides more constant gains, making the path easier for real users to follow. On SSQ scores, our method outperforms the other two because its generated paths are simple, consisting of only a circular arc and a straight line with constant gains. In contrast, the curvature gain in the TAPF method changes continuously with the potential field, and the strategy of inserting 'S'-turns in OPG causes frequent changes in the direction of the curvature gain. Such frequent directional changes in curvature gain likely caused user dizziness.

5.3. Discussion

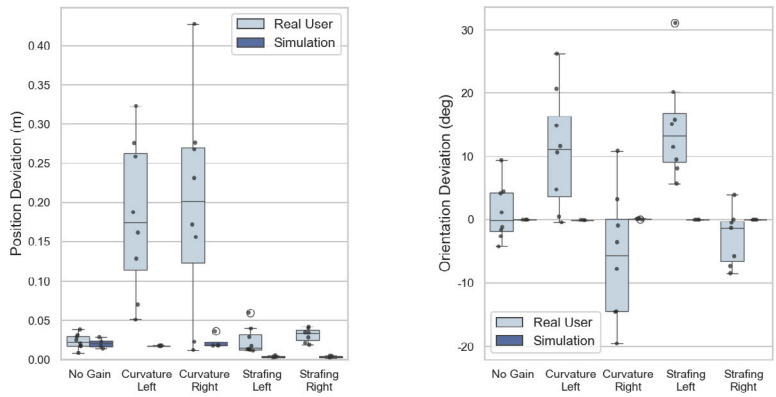
One unexpected finding is that the performance of both our method and OPG is not as good in the user study as in the simulation. For comparable simulation environment C, both OPG and our method outperform TAPF. However, in the user study, our method does not show a statistically significant advantage over TAPF, while OPG performs worse than TAPF. We attribute this performance drop to the inherent imprecision of human locomotion. Unlike simulated agents which follow an idealized trajectory, real users naturally produce small but constant deviations in their walking patterns. These errors may cause more frequent path replanning than in the simulation, which led to instability in the guidance path. Furthermore, this also triggered additional resets for OPG due to its reset in advance strategy.

To validate this hypothesis and quantify this behavioral difference, we conducted a supplementary experiment (N=8). Participants were asked to walk 5 m in a straight line toward a virtual target under five conditions: no gain, curvature gain (left/right), and strafing gain (left/right). We compared their final pose error (defined as the deviation from a calculated ideal pose) against those of simulated avatars performing the same task.

As shown in Fig. 7, the position deviations of the simulated avatars' are entirely within 0.05 m, while the orientation deviation was near 0°. In contrast, after real users walking 5 m, their position deviations are concentrated between 0.1 m and 0.3m — but the deviations for the strafing path are smaller — while the orientation deviations are between 0° and 20°. In the previous experiment, the path deviation thresholds were set at 0.2 m and 10° to ensure accurate guidance and timely adjustments. The results from this supplementary experiment suggest to some extent that real users' natural walking produces deviations that sometimes exceed our thresholds, leading to more frequent path replanning.

6. Limitation and future work

There are several limitations of our method. Firstly, our user study was limited to the complex environment to assess algorithm performance under challenging conditions. We did not conduct real user study in the obstacle-free environment, which represents a more common scenario. Future work should extend the user evaluation to include such open settings for a more comprehensive analysis. Another limitation is that our evaluation was conducted with OpenRDW. The platform's implementation of existing methods may include practical adjustments like variations in how gains are applied, which may not perfectly align with the methods' original theoretical descriptions. However, our comparison remains valid, as all compared algorithms utilize the same underlying gain implementation, ensuring a consistent



(a)Position Deviation (b)Orientation Deviation

Fig. 7. Box plots of supplementary experiment, comparing the position and orientation deviations of real users against simulation.

and fair comparison. The authors of OpenRDW also acknowledged that while reproducing exact quantitative results is impossible, their benchmark was "generally consistent to literature" and confirmed similar relative performance conclusions.

Our work can be improved in the future. We observed that real users naturally produce small but constant deviations in their walking patterns, leading to walking errors and random deviations. Future work could involve not only designing algorithms for ideal paths, but also considering their performance and robustness when subjected to user-generated deviations. Developing adaptive strategies would be a viable approach to reduce the gap between simulation and real users. Additionally, although our method provides an initial validation for integrating strafing gain, its full potential remains untapped. Future research could aim to explore more applications for strafing gain, such as its deeper integration into more complex RDW controllers for multi-user or dynamic environments.

7. Conclusion

This paper introduced a novel RDW method that mitigates the increased perceptibility and user discomfort caused when state-of-the-art trajectory planning controllers frequently switch curvature gain between opposite directions. Our main contribution is a novel path generation algorithm, integrating strafing gain with pose score guidance strategy, that ensures the direction of redirection gains constant within each guidance maneuver. The simulation results validate that our approach effectively reduces the number of resets while the user study shows lower SSQ scores, enhancing user comfort.

CRediT authorship contribution statement

Jin-Feng Li: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation. **Sen-Zhe Xu:** Writing – review & editing, Conceptualization. **Qiang Tong:** Writing – review & editing. **Peng-Hui Yuan:** Validation, Investigation. **Ling-Long Zou:** Methodology, Investigation. **Er-Xia Luo:** Investigation. **Qi Wen Gan:** Writing – review & editing. **Song-Hai Zhang:** Supervision.

Declaration of competing interest

The authors declare that there are no conflicts of interest.

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.cag.2025.104492.

Data availability

Data will be made available on request.

References

[1] Liu J, Parekh H, Al-Zayer M, Folmer E. Increasing walking in vr using redirected teleportation. In: Proceedings of the 31st annual ACM symposium on user interface software and technology. 2018, p. 521–9.
[2] Nilsson NC, Serafin S, Laursen MH, Pedersen KS, Sikström E, Nordahl R. Tapping-in-place: Increasing the naturalness of immersive walking-in-place locomotion through novel gestural input. In: 2013 IEEE symposium on 3D user interfaces. IEEE; 2013, p. 31–8.
[3] Wendt JD, Whitton MC, Brooks FP. Gud wip: Gait-understanding-driven walking-in-place. In: 2010 IEEE virtual reality conference. IEEE; 2010, p. 51–8.
[4] Warren LE, Bowman DA. User experience with semi-natural locomotion techniques in virtual reality: the case of the virtuix omni. In: Proceedings of the 5th symposium on spatial user interaction. 2017, p. 163–163.
[5] Razzaque S, Kohn Z, Whitton MC. Redirected walking. In: Eurographics 2001. 2001, http://dx.doi.org/10.2312/egs.20011036.
[6] Steinicke F, Bruder G, Jerald J, Frenz H, Lappe M. Estimation of detection thresholds for redirected walking techniques. IEEE Trans Vis Comput Graphics 2009;16(1):17–27.
[7] Razzaque S. Redirected walking. The University of North Carolina at Chapel Hill; 2005.
[8] Bachmann ER, Hodgson E, Hoffbauer C, Messinger J. Multi-user redirected walking and resetting using artificial potential fields. IEEE Trans Vis Comput Graphics 2019;25(5):2022–31.
[9] Messinger J, Hodgson E, Bachmann ER. Effects of tracking area shape and size on artificial potential field redirected walking. In: 2019 IEEE conference on virtual reality and 3D user interfaces. IEEE; 2019, p. 72–80.
[10] Thomas J, Rosenberg ES. A general reactive algorithm for redirected walking using artificial potential functions. In: 2019 IEEE conference on virtual reality and 3D user interfaces. IEEE; 2019, p. 56–62.
[11] Xu S-Z, Lv T, He G, Chen C-H, Zhang F-L, Zhang S-H. Optimal pose guided redirected walking with pose score precomputation. In: 2022 IEEE conference on virtual reality and 3D user interfaces. IEEE; 2022, p. 655–63.

[12] You C, Benda B, Rosenberg ES, Ragan E, Lok B, Thomas J. Strafing gain: Redirecting users one diagonal step at a time. In: 2022 IEEE international symposium on mixed and augmented reality. IEEE; 2022, p. 603–11.
[13] Langbehn E, Lubos P, Bruder G, Steinicke F. Bending the curve: Sensitivity to bending of curved paths and application in room-scale vr. IEEE Trans Vis Comput Graphics 2017;23(4):1389–98.
[14] Mayor J, Raya L, Bayona S, Sanchez A. Multi-technique redirected walking method. IEEE Trans Emerg Top Comput 2021;10(2):997–1008.
[15] Xu S-Z, Chen FXY, Gong R, Zhang F-L, Zhang S-H. Bird: Using bidirectional rotation gain differences to redirect users during back-and-forth head turns in walking. IEEE Trans Vis Comput Graphics 2024;30(5):2693–702.
[16] Hu L, Zhang Y, Wang R, Gao Z, Bao H, Hua W. Human sensitivity to slopes of slanted paths. In: 2019 IEEE conference on virtual reality and 3D user interfaces. IEEE; 2019, p. 984–5.
[17] Matsumoto K, Narumi T, Tanikawa T, Hirose M. Walking uphill and downhill: redirected walking in the vertical direction. In: ACM SIGGRAPH 2017 posters. 2017, p. 1–2.
[18] Matsumoto K, Langbehn E, Narumi T, Steinicke F. Detection thresholds for vertical gains in vr and drone-based telepresence systems. In: 2020 IEEE conference on virtual reality and 3D user interfaces. IEEE; 2020, p. 101–7.
[19] Cho Y-H, Min D-H, Huh J-S, Lee S-H, Yoon J-S, Lee I-K. Walking outside the box: Estimation of detection thresholds for non-forward steps. In: 2021 IEEE virtual reality and 3D user interfaces. IEEE; 2021, p. 448–54.
[20] Dong T, Gao T, Dong Y, Wang L, Hu K, Fan J. Free-rdw: A multi-user redirected walking method for supporting non-forward steps. IEEE Trans Vis Comput Graphics 2023;29(5):2315–25.
[21] Liu J-H, Ren Y-F, Gan QW, Huang K, Chen FXY, Luo E-X, Tang KY, Fu Y-Y, Fan C-W, Xu S-Z, et al. Overcoming spatial constraints in VR: A survey of redirected walking techniques. J Comput Sci Tech 2024;39(4):841–70.
[22] Chen J-J, Hung H-C, Sun Y-R, Chuang J-H. APF-S2T: Steering to target redirection walking based on artificial potential fields. IEEE Trans Vis Comput Graphics 2024;30(5):2464–73.
[23] Lee D-Y, Cho Y-H, Lee I-K. Real-time optimal planning for redirected walking using deep q-learning. In: 2019 IEEE conference on virtual reality and 3D user interfaces. IEEE; 2019, p. 63–71.
[24] Strauss RR, Ramanujan R, Becker A, Peck TC. A steering algorithm for redirected walking using reinforcement learning. IEEE Trans Vis Comput Graphics 2020;26(5):1955–63.
[25] Lee HJ, Jeon S-B, Cho Y-H, Lee I-K. Redirection strategy switching: Selective redirection controller for dynamic environment adaptation. IEEE Trans Vis Comput Graphics 2024;30(5):2474–84.
[26] Xu S-Z, Liu T-Q, Liu J-H, Zollmann S, Zhang S-H. Making resets away from targets: Poi aware redirected walking. IEEE Trans Vis Comput Graphics 2022;28(11):3778–87.
[27] Fan C-W, Xu S-Z, Yu P, Zhang F-L, Zhang S-H. Redirected walking based on historical user walking data. In: 2023 IEEE conference virtual reality and 3D user interfaces. IEEE; 2023, p. 53–62.
[28] Xu S-Z, Huang K, Fan C-W, Zhang S-H. SafeRDW: Keep VR users safe when jumping using redirected walking. In: 2024 IEEE conference virtual reality and 3D user interfaces. IEEE; 2024, p. 365–75.
[29] Luo E-X, Tong Q, Zou L-L, Liu X-T. Predicting user intention as next direction in redirected walking. In: International conference on extended reality. Springer; 2024, p. 264–80.
[30] Hirt C, Kompis Y, Holz C, Kunz A. The chaotic behavior of redirection–revisiting simulations in redirected walking. In: 2022 IEEE conference on virtual reality and 3D user interfaces. IEEE; 2022, p. 524–33.
[31] Azmandian M, Yahata R, Grechkin T, Thomas J, Rosenberg ES. Validating simulation-based evaluation of redirected walking systems. IEEE Trans Vis Comput Graphics 2022;28(5):2288–98.
[32] Li Y-J, Wang M, Steinicke F, Zhao Q. Openrdw: A redirected walking library and benchmark with multi-user, learning-based functionalities and state-of-the-art algorithms. In: 2021 IEEE international symposium on mixed and augmented reality. IEEE; 2021, p. 21–30.
[33] Kennedy RS, Lane NE, Berbaum KS, Lilienthal MG. Simulator sickness questionnaire: An enhanced method for quantifying simulator sickness. Int J Aviat Psychol 1993;3(3):203–20.
[34] Grant DA. The latin square principle in the design and analysis of psychological experiments. Psychol Bull 1948;45(5):427.